Title & Team Members

- Project title – Big Data Pipeline for Healthcare Analytics
- Names & student IDs of all group members –

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Introduction

Brief summary of the dataset and problem statement

We used a healthcare dataset containing 100,000 patient records in CSV format obtained from Kaggle. The dataset includes patient demographics, diseases, and medical conditions.
The main goal of this project is to analyze healthcare data to identify patterns such as common diseases, age-related risks, and demographic trends using Big Data technologies.

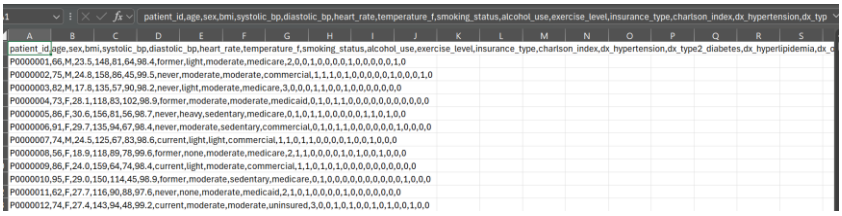
Why this problem is relevant in a Big Data context?

This problem is relevant in a Big Data context because healthcare systems generate massive amounts of data every day. Traditional tools cannot efficiently process such large datasets.
By using tools like Apache Spark and MongoDB, we can process large-scale healthcare data quickly and extract useful insights that can help in disease prediction, resource planning, and improving healthcare services.

Methodology

1. Dataset Details

   The dataset consists of 100,000 records in CSV format. Each record represents a patient and includes attributes such as age, gender, and medical conditions.
   Preprocessing steps included:
   - Removing missing values
   - Removing duplicate records
   - Converting data types where necessary



```
patient_id,age,sex,bmi,systolic_bp,diastolic_bp,heart_rate,temperature_f,smoking_status,alcohol_use,exercise_level,insurance_type,charlson_index,dx_hypertension,dx_typ
patient_id,age,sex,bmi,systolic_bp,diastolic_bp,heart_rate,temperature_f,smoking_status,alcohol_use,exercise_level,insurance_type,charlson_index,dx_hypertension,dx_type2_diabetes,dx_hyperlipidemia,dx_c
P0000001,66,M,23.5,148,81,64,98.4,former,light,moderate,medicare,2,0,0,1,0,0,0,0,1,0,0,0,0,0,1,0
P0000002,75,M,24.8,158,86,45,99.5,never,moderate,moderate,commercial,1,1,1,0,1,0,0,0,0,0,1,0,0,0,1,0
P0000003,82,M,17.8,135,57,90,98.2,never,light,moderate,medicare,3,0,0,0,1,1,0,0,1,0,0,0,0,0,0,0
P0000004,73,F,28.1,118,83,102,98.9,former,moderate,moderate,medicaid,0,1,0,1,1,1,0,0,0,0,0,0,0,0,0,0
P0000005,86,F,30.6,156,81,56,98.7,never,heavy,sedentary,medicare,0,1,0,1,1,0,0,0,0,0,1,1,0,1,0,0
P0000006,91,F,29.7,135,94,67,98.4,never,moderate,sedentary,commercial,0,1,0,1,1,1,0,0,0,0,0,0,1,0,0,0,0
P0000007,74,M,24.5,125,67,83,98.6,current,light,light,commercial,1,1,0,1,1,0,0,0,0,1,0,0,1,0,0,0
P0000008,56,F,18.9,118,89,78,99.6,former,none,moderate,medicare,2,1,1,0,0,0,0,1,0,1,0,0,0,1,0,0,0
P0000009,86,F,24.0,159,64,74,98.4,current,light,moderate,commercial,1,1,0,1,0,1,0,0,0,0,0,0,0,0,0,0,0
P0000010,95,F,29.0,150,114,45,98.9,former,moderate,sedentary,medicare,0,1,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0
P0000011,62,F,27.7,116,90,88,97.6,never,none,moderate,medicaid,2,1,0,1,0,0,0,1,1,0,0,0,0,0,0,0,0
P0000012,74,F,27.4,143,94,48,99.2,current,moderate,moderate,uninsured,3,0,0,1,0,1,0,1,0,1,0,1,0,0,1,0,0
```

Figure 1 - Raw Dataset

2. Architecture / Pipeline

The system follows a Big Data pipeline:

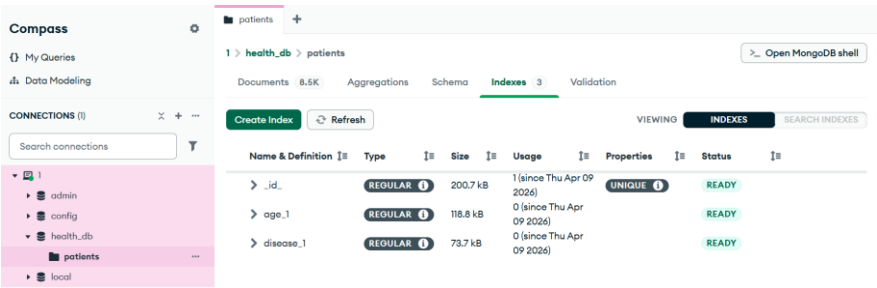
Raw Data → MongoDB (Storage) → Apache Spark (Processing) → Analysis & Visualization

- MongoDB was used to store the dataset in a flexible format
- Apache Spark was used for distributed data processing
- Spark SQL was used for querying and analysis

3. Schema Design & Indexing

MongoDB uses a document-based schema. Each patient record is stored as a JSON-like document. We used a flat document schema (each document represents one patient.

Indexes were created on important fields such as age, disease. This improved query performance by reducing search time.



Implementation

Loading and cleaning the dataset

The dataset was loaded using Pandas from a CSV file. Thereafter, data was cleaned removing null values and dropping duplicates.

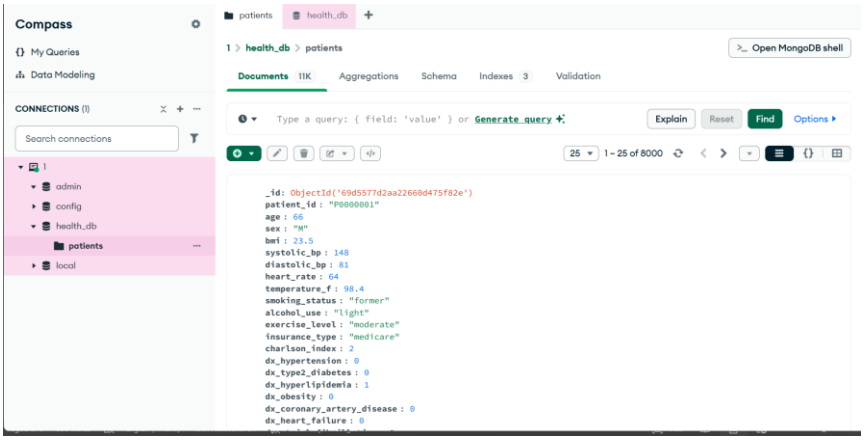
```
Preview of dataset:
  patient_id  age sex   bmi  systolic_bp  ...  dx_depression  dx_anxiety  dx_hypothyroidism dx_osteoarthritis dx_type1_diabetes
0   P0000001   66   M  23.5          148  ...              0           0                  0                 1                 0
1   P0000002   75   M  24.8          158  ...              0           0                  0                 1                 0
2   P0000003   82   M  17.8          135  ...              0           0                  0                 0                 0
3   P0000004   73   F  28.1          118  ...              0           0                  0                 0                 0
4   P0000005   86   F  30.6          156  ...              1           0                  1                 0                 0

[5 rows x 28 columns]
Data cleaned!
```

Figure 2 – Data cleaned

Storing the dataset in MongoDB

The cleaned dataset was stored in MongoDB using PyMongo.
Each row was converted into a document and inserted into a collection.



Processing and transforming the data using Apache Spark

Apache Spark (PySpark) was used to process large-scale data. DataFrames were used to perform transformations such as filtering and aggregation.

```
-----------+
|patient_id|age|sex| bmi|systolic_bp|diastolic_bp|heart_rate|temperature_f|smoking_status|alcohol_use|exercise_level|insurance_ty
pe|charlson_index|dx_hypertension|dx_type2_diabetes|dx_hyperlipidemia|dx_obesity|dx_coronary_artery_disease|dx_heart_failure|dx_a
trial_fibrillation|dx_chronic_kidney_disease|dx_copd|dx_asthma|dx_depression|dx_anxiety|dx_hypothyroidism|dx_osteoarthritis|dx_ty
pe1_diabetes|
+---------+---+---+----+-----------+------------+----------+-------------+--------------+-----------+--------------+-----------
--+--------------+---------------+---------------+-----------------+---------+-----------------------+---------------+----
---------------+-------------------------+-------+---------+-------------+---------+----------------+----------------+----
-----------+
| P0000001| 66|  M|23.5|       148|          81|        64|         98.4|        former|      light|      moderate|      medica
re|            2|             0|              0|                 1|        0|                       0|              0|              0|
          0|                     1|      0|       0|          0|       0|               0|              1|
| P0000002| 75|  M|24.8|       158|          86|        45|         99.5|         never|   moderate|      moderate|    commerci
al|            1|             1|              1|                 0|        1|                       0|              0|              0|
          0|                     0|      0|       1|          0|       0|               0|              1|
| P0000003| 82|  M|17.8|       135|          57|        90|         98.2|         never|      light|      moderate|      medica
re|            3|             0|              0|                 0|        1|                       1|              0|              0|
          0|                     1|      0|       0|          0|       0|               0|              0|
| P0000004| 73|  F|28.1|       118|          83|       102|         98.9|        former|   moderate|      moderate|      medica
id|            0|             1|              0|                 1|        1|                       0|              0|              0|
          0|                     0|      0|       0|          0|       0|               0|              0|
```

```
+---------------+-----+
|dx_hypertension|count|
+---------------+-----+
|              1|67685|
|              0|32315|
+---------------+-----+

Patients older than 50:
+----------+---+---+----+-----------+------------+----------+-------------+--------------+-----------+--------------+-----------
--+--------------+---------------+---------------+-----------------+---------+-----------------------+---------------+----
-----------------+-------------------------+-------+---------+-------------+---------+----------------+----------------+----
-----------+
|patient_id|age|sex| bmi|systolic_bp|diastolic_bp|heart_rate|temperature_f|smoking_status|alcohol_use|exercise_level|insurance_ty
pe|charlson_index|dx_hypertension|dx_type2_diabetes|dx_hyperlipidemia|dx_obesity|dx_coronary_artery_disease|dx_heart_failure|dx_a
trial_fibrillation|dx_chronic_kidney_disease|dx_copd|dx_asthma|dx_depression|dx_anxiety|dx_hypothyroidism|dx_osteoarthritis|dx_ty
pe1_diabetes|
+----------+---+---+----+-----------+------------+----------+-------------+--------------+-----------+--------------+-----------
--+--------------+---------------+---------------+-----------------+---------+-----------------------+---------------+----
-----------------+-------------------------+-------+---------+-------------+---------+----------------+----------------+----
-----------+
| P0000001| 66|  M|23.5|       148|          81|        64|         98.4|        former|      light|      moderate|      medica
```

Several Spark transformations were performed to analyze the dataset efficiently:

- Filtering patients older than 50
- Grouping data by diseases
- Aggregating totals for different conditions

These operations were executed in parallel using Spark, improving performance.

Spark SQL

Spark SQL was used to run SQL queries on the dataset. This allowed easier analysis using SQL syntax.

```
Spark SQL result:
+---------------+--------+
|dx_hypertension|count(1)|
+---------------+--------+
|              1|   67685|
|              0|   32315|
+---------------+--------+

Disease totals:
+------------+--------------+-------+
|hypertension|type2_diabetes|obesity|
+------------+--------------+-------+
|       67685|         15750|  60309|
+------------+--------------+-------+

Heart failure by age:
+---+---------------------+
|age|avg(dx_heart_failure)|
+---+---------------------+
| 31|  0.01762114537444934|
| 85|   0.0345005149330587|
| 65|  0.03167420814479638|
| 53| 0.023034154090548053|
| 78|  0.02964824120603015|
```

Visualization

Multiple visualizations were created to better understand the dataset.
While initial visualizations were created using Pandas and Seaborn, additional aggregations were performed using Apache Spark to demonstrate distributed data processing. The results were then converted to Pandas DataFrames for visualization.

Results & Insights

Key findings from data analysis

The analysis revealed several key insights such as:

- Certain diseases are more common in specific age groups
- Some conditions appear frequently across all demographics
- Data patterns can help in early detection of health risks

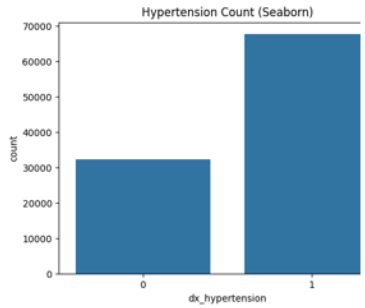
Also, the following shows the execution plan for the hypertension aggregation query:

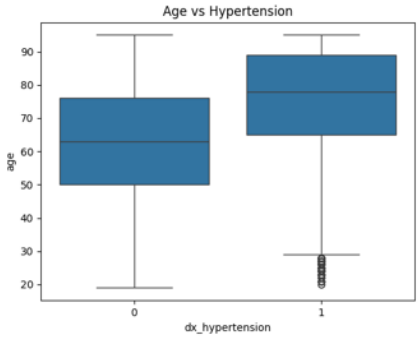
```
== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- HashAggregate(keys=[dx_hypertension#30], functions=[count(1)], output=[dx_hypertension#30, count(1)#325L])
   +- Exchange hashpartitioning(dx_hypertension#30, 200), ENSURE_REQUIREMENTS, [plan_id=144]
      +- HashAggregate(keys=[dx_hypertension#30], functions=[partial_count(1)], output=[dx_hypertension#30, count#327L])
         +- FileScan csv [dx_hypertension#30] Batched: false, DataFilters: [], Format: CSV, Location: InMemoryFileIndex(1 paths)[file:/C:/Users/anjel/Documents/HAMK/YEAR 3/Big Data/patients.csv],
 PartitionFilters: [], PushedFilters: [], ReadSchema: struct<dx_hypertension:int>
```

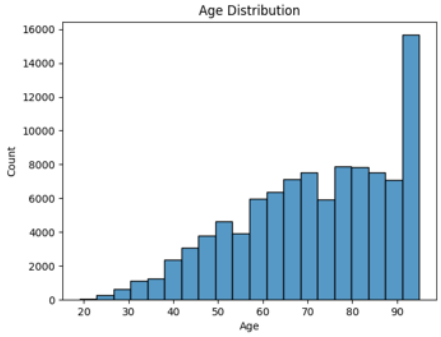
Performance evaluation (e.g., query execution time comparison)

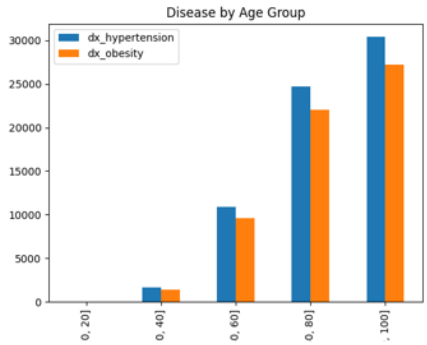
Spark SQL queries performed faster compared to traditional methods due to distributed processing.
MongoDB indexing significantly improved query performance.

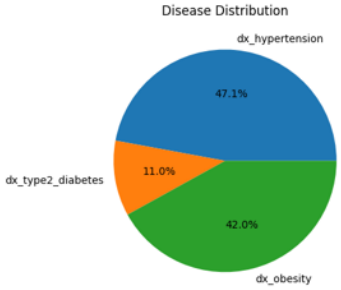
Visualization of results (tables, graphs, charts)

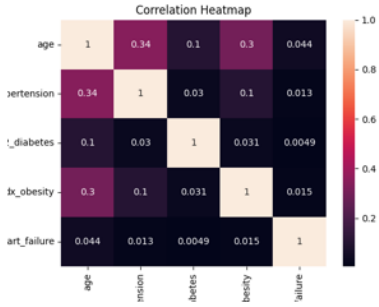












Conclusion & Future Work

Challenges:

- Handling large datasets
- Data cleaning complexity
- Understanding Spark architecture

Future Improvements:

- Use real-time streaming data
- Improve schema design
- Build a dashboard for visualization